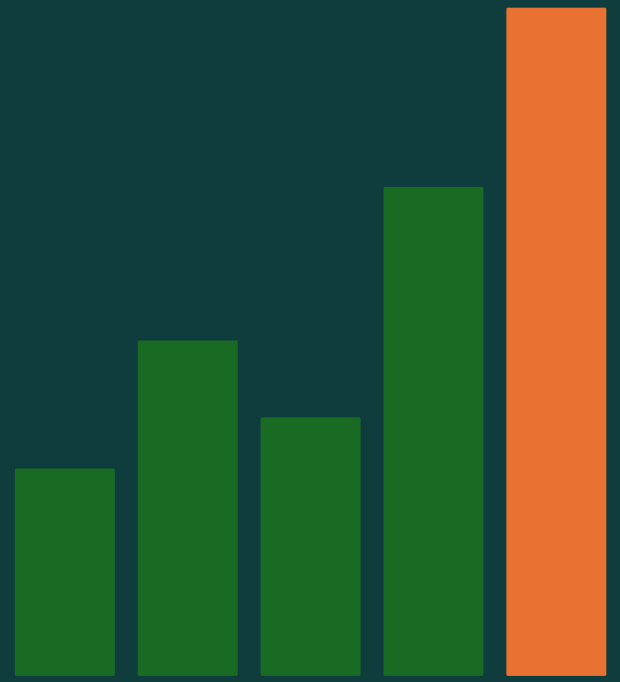
# Smart Retail Analytics & Sales Forecasting System

Major Project: Data Analytics + Machine Learning

Presented by: K.Keerthika | 620324104046

BE/ CSE, Bharathiyar Institute of Engineering for Women.



# Problem Statement

## The problem

Retailers collect large volumes of transaction data, but turning it into clear decisions about products, customers and stock is hard without a structured analytics workflow.

*Scope: sales analysis, SQL database, interactive dashboard and a baseline forecast.*

1. Which products and categories earn the most revenue and profit?
2. Which months, regions or channels perform best?
3. Who are the most valuable and repeat customers?
4. Which products show declining sales or stock issues?
5. What sales can we expect in the next period?

# Objectives



## Clean & Analyse

Clean transactions with Pandas and explore sales, product and customer patterns.



## Store & Query

Build a SQL database and answer business questions with verified queries.



## Visualise & Predict

Deliver an interactive dashboard and a baseline sales forecast.

**Target users:** store managers, merchandising teams and business owners

# Dataset

## Data source

Public retail / e-commerce dataset, or a clearly labelled synthetic dataset.

Synthetic Retail Sales Dataset (sales_data.csv) github.com/keerthikaBE/smart-retail-analytics

A data dictionary describes every column and its meaning.

- Order ID
- Order Date
- Customer ID
- Product ID
- Category
- Quantity
- Unit Price
- Discount
- Region
- Sales Channel
- Cost / Profit*

*If available. Only columns present in the dataset are used; no invented fields or unsupported profit claims.

# Technology Stack



PROGRAMMING

**Python**

Logic, automation, analysis



DATA PROCESSING

**Pandas, NumPy**

Clean, transform, calculate metrics



DATABASE

**SQLite or MySQL**

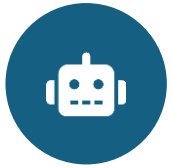
Store data, answer business queries



VISUALIZATION

**Power BI or Streamlit**

Interactive dashboard with filters



MACHINE LEARNING

**scikit-learn**

Baseline forecasting model



DELIVERY

**GitHub + README**

Version control and presentation

# Project Workflow




| 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|
| **Define problem** | **Collect data** | **Clean & transform** | **Analyse & query** | **Dashboard & model** | **Recommend** |
| Scope and 5-8 business questions | Dataset and data dictionary | Pandas cleaning, revenue fields | EDA, SQL checked against Python | KPIs, 4+ views, baseline forecast | 3-5 evidence-based actions |

*Reliable analysis and dashboard first. Machine learning is added only when the data supports it.*

# Data Processing: Python & SQL

## Python (Pandas, NumPy)

- Inspect types, missing values, duplicates
- Standardise dates and categories
- Create revenue and profit fields
- Document every cleaning decision
- EDA: monthly trends, categories, customer frequency, outliers

## SQL database

- Create tables and import cleaned data
- SELECT, WHERE, GROUP BY, JOIN, aggregates
- Answer the business questions with queries
- Verify SQL results against Python
- Keep schema and query file in the repo

# Interactive Dashboard

### KPI cards

Revenue, orders, avg order value, margin

### Monthly trends

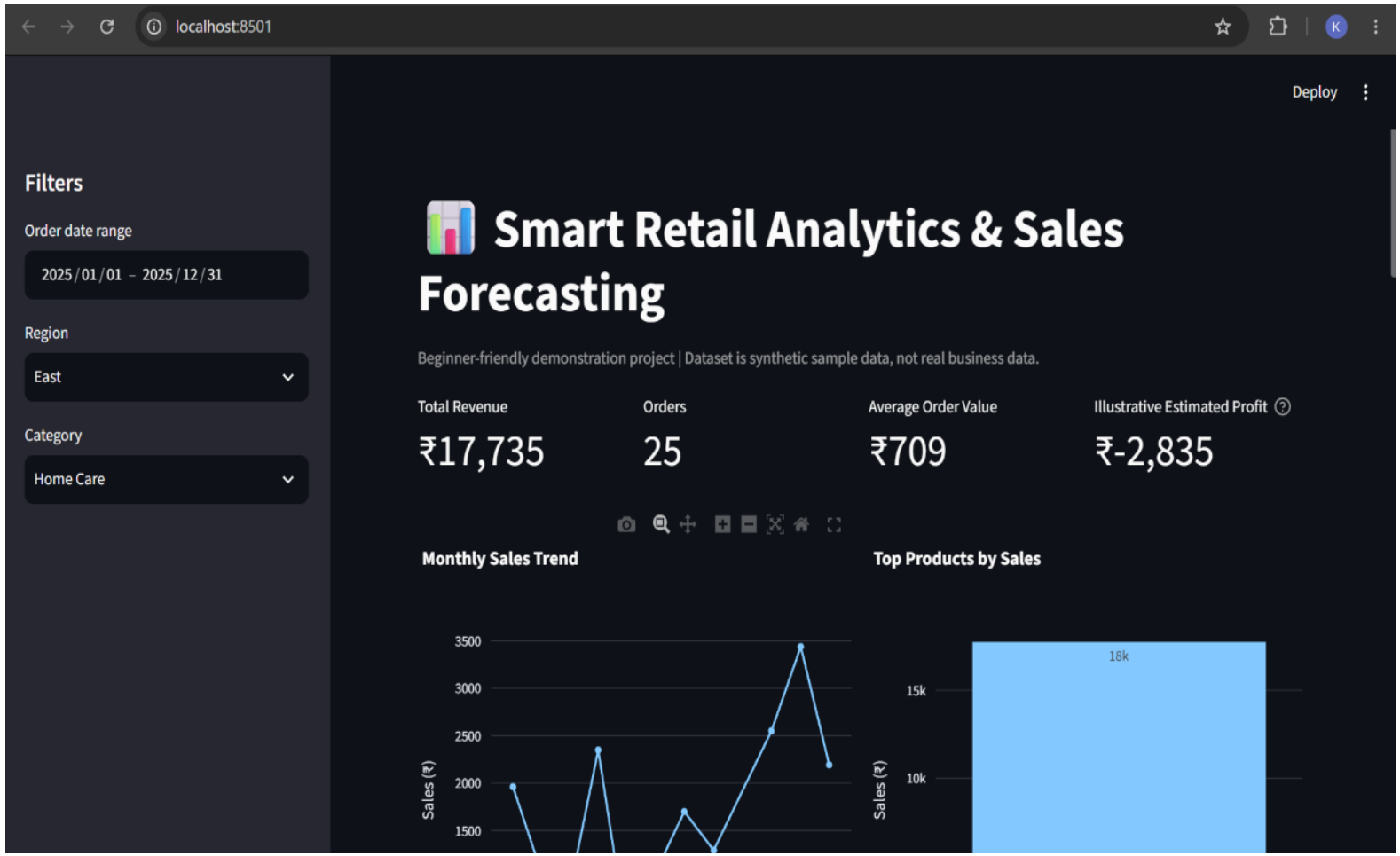
Sales over time

### Top products

Best products and categories

### Filters

Region and category slicers



# Forecasting Model & Validation

## Approach

- Objective: forecast weekly or monthly sales (or churn, if a valid label exists)
- Time-aware train / test split
- Compare against a simple baseline
- Evaluate with suitable metrics (e.g. MAE, RMSE)
- Explain limits and data-leakage risks

## Results

| Model | MAE | RMSE |
|---|---|---|
| Baseline | ₹5,434 | ₹5,499 |
| scikit-learn model | ₹10,506 | ₹11,099 |

*Machine learning is an additional component. Report the numbers honestly, including what the model does not prove.*

# Business Recommendations

| Finding area | Recommendation | Expected benefit |
|---|---|---|
| **Top products & categories** | Keep Cooking Oil (24%) and Grocery (31%) stocked; review pricing of Cooking Oil, Detergent, Soap; promote Beverages | Fewer stock-outs; better margin |
| **Declining sales & stock** | Oct–Nov weakest; cut Shampoo and Coffee replenishment, use targeted offers | Lower overstock cost |
| **Best months, regions, channels** | Prepare stock and online campaigns for Aug peak; run local promotions in West (-36%) | Capture peak demand; recover West |

*Each recommendation should link to a chart or metric and a business benefit.*

# Limitations & Future Scope

## Limitations

- Results depend on dataset quality and coverage
- No profit claims without cost data
- Forecast is a baseline, not proof of cause
- Assumptions and leakage risks are documented

## Future scope

- Churn prediction when a valid label exists
- Richer forecasting models
- Automated data refresh pipeline
- Deploy the dashboard online

# Thank You

Any questions?

GitHub: github.com/keerthikaBE/smart-retail-analytics

Email: keerthibe2007@gmail.com

